

CS 2601 Linear and Convex Optimization

12. Newton's method for equality constrained problems

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Outline

- Equality constrained convex QP
- Newton's method with feasible start

Equality constrained convex QP

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{g}^T \mathbf{x} + c \\ \text{s.t.} \quad & \mathbf{A} \mathbf{x} = \mathbf{b} \end{aligned} \tag{QP}$$

where $\mathbf{Q} \in \mathbb{R}^n$, $\mathbf{Q} \succeq \mathbf{O}$, $\mathbf{A} \in \mathbb{R}^{k \times n}$, $\text{rank} \mathbf{A} = k$.

- The Lagrange condition $\nabla \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = \mathbf{0}$ gives the **KKT system** of the problem,

$$\begin{cases} \nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = \mathbf{Q} \mathbf{x}^* + \mathbf{g} + \mathbf{A}^T \boldsymbol{\lambda}^* = \mathbf{0} \\ \nabla_{\boldsymbol{\lambda}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = \mathbf{A} \mathbf{x}^* - \mathbf{b} = \mathbf{0} \end{cases} \quad \text{or} \quad \begin{bmatrix} \mathbf{Q} & \mathbf{A}^T \\ \mathbf{A} & \mathbf{O} \end{bmatrix} \begin{bmatrix} \mathbf{x}^* \\ \boldsymbol{\lambda}^* \end{bmatrix} = \begin{bmatrix} -\mathbf{g} \\ \mathbf{b} \end{bmatrix}$$

The coefficient matrix $\mathbf{K} = \begin{bmatrix} \mathbf{Q} & \mathbf{A}^T \\ \mathbf{A} & \mathbf{O} \end{bmatrix}$ is called the **KKT matrix**.

- Any solution to the KKT system gives an optimal solution $\mathbf{x}^*$ with corresponding Lagrange multiplier $\boldsymbol{\lambda}^*$.

Solutions to KKT system

Since A has full column rank,

$$\# \text{ solutions to KKT system} = \# \text{ minima}$$

Suppose $Ax_0 = b$ and $v \neq 0$ is a feasible direction, i.e. $v \in \text{Null}(A)$. Consider

$$\phi(t) = f(x_0 + tv) = f(x_0) + (g^T v + x_0^T Qv)t + \left(\frac{1}{2}v^T Qv\right)t^2$$

Three cases for the number of minima of $\phi(t)$

1. unique minimum

$$v^T Qv > 0 \iff Qv \neq 0$$

2. infinitely many minima

$$v^T Qv = 0, g^T v + x_0^T Qv = 0 \iff Qv = 0, g^T v = 0$$

3. no minimum

$$v^T Qv = 0, g^T v + x_0^T Qv \neq 0 \iff Qv = 0, g^T v \neq 0$$

Solutions to KKT system

The above three cases correspond to the three cases for the solutions to the KKT system.

1. unique solution (proof on slides 6-8)

$$\text{Null}(\mathbf{A}) \cap \text{Null}(\mathbf{Q}) = \{\mathbf{0}\}$$

2. infinitely many solutions

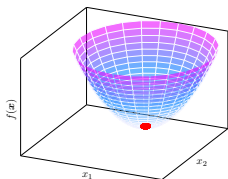
$$\text{Null}(\mathbf{A}) \cap \text{Null}(\mathbf{Q}) \neq \{\mathbf{0}\} \text{ and } \mathbf{g} \perp \text{Null}(\mathbf{A}) \cap \text{Null}(\mathbf{Q})$$

3. no solution

$$\text{Null}(\mathbf{A}) \cap \text{Null}(\mathbf{Q}) \neq \{\mathbf{0}\} \text{ and } \mathbf{g} \not\perp \text{Null}(\mathbf{A}) \cap \text{Null}(\mathbf{Q})$$

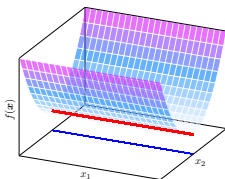
Note. When $\mathbf{A} = \mathbf{O}$, this reduces to the unconstrained case.

Unconstrained vs constrained problems



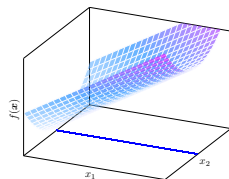
$$Q \succ 0$$

unique solution



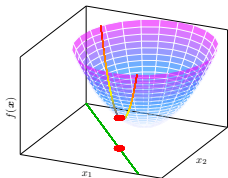
$$g \perp \text{Null}(Q) \neq \{0\}$$

infinitely many solutions



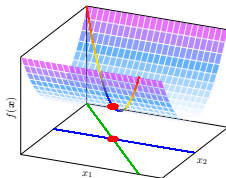
$$g \not\perp \text{Null}(Q) \neq \{0\}$$

no solution



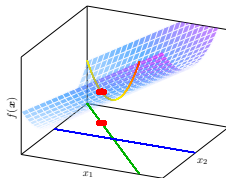
$$Q \succ 0$$

unique solution



$$\text{Null}(Q) \cap \text{Null}(A) = \{0\}$$

unique solution



$$\text{Null}(Q) \cap \text{Null}(A) = \{0\}$$

unique solution

Null space of KKT matrix

Lemma.

$$\text{Null}(\mathbf{K}) = \{(\mathbf{v}, \mathbf{0}) : \mathbf{v} \in \text{Null}(\mathbf{A}) \cap \text{Null}(\mathbf{Q})\}$$

Proof. Denote the RHS by S .

1. $S \subset \text{Null}(\mathbf{K})$.

$$(\mathbf{v}, \mathbf{0}) \in S \implies \begin{bmatrix} \mathbf{Q} & \mathbf{A}^T \\ \mathbf{A} & \mathbf{O} \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{Q}\mathbf{v} \\ \mathbf{A}\mathbf{v} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} \implies (\mathbf{v}, \mathbf{0}) \in \text{Null}(\mathbf{K})$$

2. $\text{Null}(\mathbf{K}) \subset S$.

► Let $(\mathbf{v}, \boldsymbol{\lambda}) \in \text{Null}(\mathbf{K})$, i.e. $\mathbf{Q}\mathbf{v} + \mathbf{A}^T\boldsymbol{\lambda} = \mathbf{0}$, $\mathbf{A}\mathbf{v} = \mathbf{0}$. Then

$$\mathbf{v}^T \mathbf{Q} \mathbf{v} = \mathbf{v}^T (-\mathbf{A}^T \boldsymbol{\lambda}) = -(\mathbf{A}\mathbf{v})^T \boldsymbol{\lambda} = -\mathbf{0}^T \boldsymbol{\lambda} = 0$$

► Since $\mathbf{Q} \succeq \mathbf{O}$, we have $\mathbf{Q}\mathbf{v} = \mathbf{0}$

Proof. Let $\mathbf{Q} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^T$. Then $\mathbf{Q} = \mathbf{B}^T \mathbf{B}$ with $\mathbf{B} = \boldsymbol{\Lambda}^{\frac{1}{2}} \mathbf{U}^T$. Thus $\mathbf{v}^T \mathbf{Q} \mathbf{v} = \|\mathbf{B}\mathbf{v}\|^2 = 0 \implies \mathbf{B}\mathbf{v} = \mathbf{0} \implies \mathbf{Q}\mathbf{v} = \mathbf{B}^T \mathbf{B}\mathbf{v} = \mathbf{0}$.

► $\mathbf{v} \in \text{Null}(\mathbf{A}) \cap \text{Null}(\mathbf{Q})$ and $\mathbf{A}^T \boldsymbol{\lambda} = \mathbf{0}$.

► Since $\mathbf{A}^T$ has full column rank, $\boldsymbol{\lambda} = \mathbf{0}$. Thus $\text{Null}(\mathbf{K}) \subset S$.

Nonsingularity of KKT matrix (case 1)

If the KKT matrix $\mathbf{K}$ is nonsingular, then the KKT system has a unique solution, which is optimal.

Recall $\mathbf{Q} \succeq \mathbf{O}$ and $\text{rank } \mathbf{A} = k$. The following conditions are equivalent

1. $\mathbf{K}$ is nonsingular
2. $\text{Null}(\mathbf{Q}) \cap \text{Null}(\mathbf{A}) = \{\mathbf{0}\}$, i.e. $\mathbf{Q}$ and $\mathbf{A}$ have no nontrivial common nullspace, i.e. $\mathbf{A}\mathbf{v} = \mathbf{0}$, $\mathbf{Q}\mathbf{v} = \mathbf{0}$ only have the trivial solution $\mathbf{v} = \mathbf{0}$.
3. $\mathbf{A}\mathbf{v} = \mathbf{0}, \mathbf{v} \neq \mathbf{0} \implies \mathbf{v}^T \mathbf{Q} \mathbf{v} > 0$, i.e. $\mathbf{Q}$ is positive definite on the nullspace of $\mathbf{A}$.
4. $\mathbf{F}^T \mathbf{Q} \mathbf{F} \succ \mathbf{O}$ for any $\mathbf{F} \in \mathbb{R}^{n \times (n-k)}$ s.t. $\text{Range}(\mathbf{F}) = \text{Null}(\mathbf{A})$, i.e. the columns of $\mathbf{F}$ are linearly independent solutions of $\mathbf{A}\mathbf{v} = \mathbf{0}$.

In particular, if $\mathbf{Q} \succ \mathbf{O}$, then $\mathbf{K}$ is nonsingular (by 3).

Proof

We show $1 \iff 2 \iff 3 \iff 4$.

- $(1 \iff 2)$. By the lemma on slide 6,

$$K \text{ nonsingular} \iff \text{Null}(K) = \{\mathbf{0}\} \iff \text{Null}(A) \cap \text{Null}(Q) = \{\mathbf{0}\}$$

- $(2 \iff 3)$. Since $Q \succeq O$, $\mathbf{v}^T Q \mathbf{v} = 0$ iff $Q \mathbf{v} = \mathbf{0}$ (proof on slide 6).

$$\begin{aligned} 2 &\iff (A \mathbf{v} = \mathbf{0} \text{ and } Q \mathbf{v} = \mathbf{0} \implies \mathbf{v} = \mathbf{0}) \\ &\iff (A \mathbf{v} = \mathbf{0} \text{ and } \mathbf{v} \neq \mathbf{0} \implies Q \mathbf{v} \neq \mathbf{0}) \iff 3 \end{aligned}$$

- $(3 \iff 4)$. Note $\mathbf{x} \in \text{Null}(A)$ iff $\mathbf{v} = F \mathbf{z}$, and $F \mathbf{z} \neq \mathbf{0}$ iff $\mathbf{z} \neq \mathbf{0}$,

$$\begin{aligned} 3 &\iff (\mathbf{v}^T Q \mathbf{v} > 0 \text{ if } A \mathbf{v} = \mathbf{0}, \mathbf{v} \neq \mathbf{0}) \\ &\iff (\mathbf{v}^T Q \mathbf{v} > 0 \text{ if } \mathbf{v} = F \mathbf{z}, \mathbf{z} \neq \mathbf{0}) \\ &\iff (\mathbf{z}^T F^T Q F \mathbf{z} > 0 \text{ if } \mathbf{z} \neq \mathbf{0}) \\ &\iff F^T Q F \succ O \end{aligned}$$

Example

$$\begin{array}{ll}\min_{x_1, x_2} & f(x_1, x_2) = \frac{1}{2}x_2^2 \\ \text{s.t.} & x_1 + 2x_2 = b\end{array}$$

Trivial with solution $x_1^* = b, x_2^* = 0$.

But let's check the condition on slide 7.

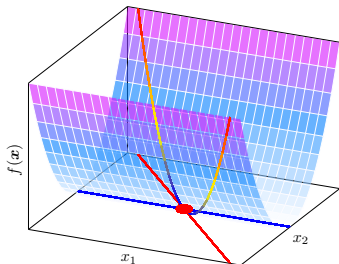
$$\mathbf{Q} = \text{diag}\{0, 1\}, \quad \mathbf{A} = [1, 2]$$

Let $\mathbf{F} = [2, -1]^T$. Then $\text{Range}(\mathbf{F}) = \text{Null}(\mathbf{A})$, and

$$\mathbf{F}^T \mathbf{Q} \mathbf{F} = [1] \succ \mathbf{0}$$

By 4 of slide 7, the KKT matrix is nonsingular, so $\exists$ a unique solution.

Note. The unconstrained problem $\min_{\mathbf{x}} f(\mathbf{x})$ has infinitely many solutions. But this does not prevent the constrained problem from having a unique solution, as $\mathbf{Q} \succ \mathbf{0}$ on $\text{Null}(\mathbf{A})$ (see 3 on slide 7).



Unsolvable case

Example.

$$\begin{array}{ll}\min_{x_1, x_2} & f(x_1, x_2) = \frac{1}{2}x_2^2 + x_1 \\ \text{s.t.} & x_2 = 0\end{array}$$

This is a convex QP with

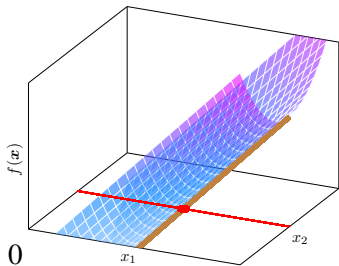
$$\mathbf{Q} = \text{diag}\{0, 1\}, \mathbf{g} = (1, 0), \mathbf{A} = [0, 1], b = 0$$

The KKT system is

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \lambda \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

which has no solution, since $0 \cdot x_1 + 0 \cdot x_2 + 0 \cdot \lambda \neq -1$.

Note $f^* = -\infty$, and $\mathbf{e}_1 = (1, 0) \in \text{Null}(\mathbf{A}) \cap \text{Null}(\mathbf{Q})$ and $\mathbf{g}^T \mathbf{e}_1 \neq 0$, i.e. there is a feasible direction along which the linear term dominates.



Outline

- Equality constrained convex QP
- Newton's method with feasible start

Newton direction

Consider the convex problem with equality constraints,

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{A}\mathbf{x} = \mathbf{b}\end{array}$$

The **Newton direction** at $\mathbf{x}_k$ is the solution to the approximation

$$\begin{array}{ll}\min_{\mathbf{v}} & \hat{f}(\mathbf{x}_k + \mathbf{v}) = f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T \mathbf{v} + \frac{1}{2} \mathbf{v}^T \nabla^2 f(\mathbf{x}_k) \mathbf{v} \\ \text{s.t.} & \mathbf{A}(\mathbf{x}_k + \mathbf{v}) = \mathbf{b}\end{array}$$

For **feasible** $\mathbf{x}_k$, the constraints become $\mathbf{A}\mathbf{v} = \mathbf{0}$. The KKT system is

$$\begin{bmatrix} \nabla^2 f(\mathbf{x}_k) & \mathbf{A}^T \\ \mathbf{A} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} -\nabla f(\mathbf{x}_k) \\ \mathbf{0} \end{bmatrix}$$

We will assume the KKT matrix is always nonsingular (cf. slide 7).

Newton direction (cont'd)

The Newton direction is a descent direction when $\mathbf{v} \neq \mathbf{0}$. In fact,

$$\left. \frac{d}{dt} f(\mathbf{x}_k + t\mathbf{v}) \right|_{t=0} = \nabla f(\mathbf{x}_k)^T \mathbf{v}$$

by the first KKT equation,

$$\begin{aligned} &= -(\nabla^2 f(\mathbf{x}_k) \mathbf{v} + \mathbf{A}^T \boldsymbol{\lambda})^T \mathbf{v} \\ &= -\mathbf{v}^T \nabla^2 f(\mathbf{x}_k) \mathbf{v} - \boldsymbol{\lambda}^T \mathbf{A} \mathbf{v} \end{aligned}$$

by the second KKT equation,

$$= -\mathbf{v}^T \nabla^2 f(\mathbf{x}_k) \mathbf{v}$$

Since $\mathbf{v} \in \text{Null}(\mathbf{A})$, 3 of slide 7 implies $\mathbf{v}^T \nabla^2 f(\mathbf{x}_k) \mathbf{v} > 0$ and hence

$$\left. \frac{d}{dt} f(\mathbf{x}_k + t\mathbf{v}) \right|_{t=0} < 0$$

Linearization of Lagrange condition

The Lagrange condition for the original problem is

$$\begin{aligned}\nabla f(\mathbf{x}^*) + \mathbf{A}^T \boldsymbol{\lambda} &= \mathbf{0} \\ \mathbf{A} \mathbf{x}^* &= \mathbf{b}\end{aligned}$$

Let $\mathbf{x}^* = \mathbf{x}_k + \mathbf{v}$ for a feasible $\mathbf{x}_k$. The above equations become

$$\begin{aligned}\nabla f(\mathbf{x}_k + \mathbf{v}) + \mathbf{A}^T \boldsymbol{\lambda} &= \mathbf{0} \\ \mathbf{A} \mathbf{v} &= \mathbf{0}\end{aligned}$$

If we linearize the first equation by $\nabla f(\mathbf{x}_k + \mathbf{v}) \approx \nabla f(\mathbf{x}_k) + \nabla^2 f(\mathbf{x}_k) \mathbf{v}$, then

$$\begin{aligned}\nabla f(\mathbf{x}_k) + \nabla^2 f(\mathbf{x}_k) \mathbf{v} + \mathbf{A}^T \boldsymbol{\lambda} &\approx \mathbf{0} \\ \mathbf{A} \mathbf{v} &= \mathbf{0}\end{aligned}$$

Newton's method can be considered an iterative method for solving the Lagrange condition.

(Damped) Newton's method with feasible start

- 1: initialization $\mathbf{x} \leftarrow \mathbf{x}_0 \in X$ $\triangleright \mathbf{x}_0$ is feasible, i.e. $A\mathbf{x}_0 = \mathbf{b}$
- 2: **repeat**
- 3: **Compute Newton's direction $\mathbf{v}$ by solving**

$$\begin{bmatrix} \nabla^2 f(\mathbf{x}) & A^T \\ A & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \lambda \end{bmatrix} = \begin{bmatrix} -\nabla f(\mathbf{x}) \\ \mathbf{0} \end{bmatrix}$$

- 4: $t \leftarrow 1$ $\triangleright$ backtracking line search on lines 4-7
- 5: **while** $f(\mathbf{x} + t\mathbf{v}) > f(\mathbf{x}) + \alpha t \nabla f(\mathbf{x})^T \mathbf{v}$ **do**
- 6: $t \leftarrow \beta t$
- 7: **end while**
- 8: $\mathbf{x} \leftarrow \mathbf{x} + t\mathbf{v}$
- 9: **until** $\|\mathbf{v}\| \leq \delta$
- 10: **return** $\mathbf{x}$

Note. We **cannot** use $\|\nabla f(\mathbf{x})\| \leq \delta$ as stopping criterion now, as $\nabla f(\mathbf{x}^*) = \mathbf{0}$ no longer holds in general. [BV] uses $\sqrt{\mathbf{v}^T \nabla^2 f(\mathbf{x}) \mathbf{v}} \leq \delta$.

Note. This is called a **feasible descent method**, since all $\mathbf{x}_k$ are feasible and $f(\mathbf{x}_{k+1}) < f(\mathbf{x}_k)$ unless $\mathbf{x}_k$ is optimal.

Convergence analysis

Let $\mathbf{F} \in \mathbb{R}^{n \times (n-k)}$ be a matrix whose columns are linearly independent solutions to $\mathbf{A}\mathbf{v} = \mathbf{0}$. Fix a feasible $\tilde{\mathbf{x}} \in X$. Every $\mathbf{x} \in X$ has a unique representation of the form $\mathbf{x} = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}$,

$$X = \{\mathbf{x} : \mathbf{A}\mathbf{x} = \mathbf{b}\} = \{\tilde{\mathbf{x}} + \mathbf{F}\mathbf{z} : \mathbf{z} \in \mathbb{R}^{n-k}\}$$

Constrained problem reduces to unconstrained problem by $\mathbf{x} = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}$,

$$(C) : \begin{cases} \min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{A}\mathbf{x} = \mathbf{b} \end{cases} \iff (U) : \min_{\mathbf{z}} g(\mathbf{z}) = f(\tilde{\mathbf{x}} + \mathbf{F}\mathbf{z})$$

Applying Newton's method to (C) with initial point $\mathbf{x}_0 = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}_0$ is equivalent to applying Newton's method to (U) with initial point $\mathbf{z}_0$:

If $\{\mathbf{x}_k\}$ and $\{\mathbf{z}_k\}$ are the iterates for (C) and (U), respectively, then

$$\mathbf{x}_k = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}_k, \quad \forall k$$

Newton's method has same convergence properties for (C) and (U). 16

Proof

We only need to show $\mathbf{x}_1 = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}_1$ and then use induction.

The Newton direction $\mathbf{v}_0$ at $\mathbf{x}_0$ for (C) is the unique solution to

$$\begin{aligned} \min_{\mathbf{v}} \quad & \hat{f}(\mathbf{x}_0 + \mathbf{v}) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T \mathbf{v} + \frac{1}{2} \mathbf{v}^T \nabla^2 f(\mathbf{x}_0) \mathbf{v} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{v} = \mathbf{0} \end{aligned}$$

The Newton direction $\mathbf{u}_0$ at $\mathbf{z}_0$ for (U) is the unique (why?) solution to

$$\begin{aligned} \min_{\mathbf{u}} \quad & \hat{g}(\mathbf{z}_0 + \mathbf{u}) = g(\mathbf{z}_0) + \nabla g(\mathbf{z}_0)^T \mathbf{u} + \frac{1}{2} \mathbf{u}^T \nabla^2 g(\mathbf{z}_0) \mathbf{u} \\ & = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T \mathbf{F}\mathbf{u} + \frac{1}{2} \mathbf{u}^T \mathbf{F}^T \nabla^2 f(\mathbf{x}_0) \mathbf{F}\mathbf{u} \end{aligned}$$

where, by the chain rule, $\nabla g(\mathbf{z}_0) = \mathbf{F}^T \nabla f(\mathbf{x}_0)$, $\nabla^2 g(\mathbf{z}_0) = \mathbf{F}^T \nabla^2 f(\mathbf{x}_0) \mathbf{F}$.

The two problems are related by $\mathbf{v} = \mathbf{F}\mathbf{u}$. In particular, $\mathbf{v}_0 = \mathbf{F}\mathbf{u}_0$.

Proof (cont'd)

Now we show backtracking line search gives the same step size t_0 for both (C) and (U).

Note

$$f(\mathbf{x}_0) = f(\tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}_0) = g(\mathbf{z}_0)$$

$$f(\mathbf{x}_0 + t\mathbf{v}_0) = f(\tilde{\mathbf{x}} + \mathbf{F}(\mathbf{z}_0 + t\mathbf{u}_0)) = g(\mathbf{z}_0 + t\mathbf{u}_0)$$

$$\nabla f(\mathbf{x}_0)^T \mathbf{v}_0 = \nabla f(\mathbf{x}_0)^T \mathbf{F}\mathbf{u}_0 = \nabla g(\mathbf{z}_0)^T \mathbf{u}_0$$

Thus the test condition in backtracking line search for f

$$f(\mathbf{x}_0 + t\mathbf{v}_0) > f(\mathbf{x}_0) + \alpha t \nabla f(\mathbf{x}_0)^T \mathbf{v}_0$$

is exactly the same as that for g ,

$$g(\mathbf{z}_0 + t\mathbf{u}_0) > g(\mathbf{z}_0) + \alpha t \nabla g(\mathbf{z}_0)^T \mathbf{u}_0$$

Finally,

$$\mathbf{x}_1 = \mathbf{x}_0 + t_0\mathbf{v}_0 = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}_0 + t_0\mathbf{F}\mathbf{u}_0 = \tilde{\mathbf{x}} + \mathbf{F}(\mathbf{z}_0 + t_0\mathbf{u}_0) = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}_1$$